

Targeted Deanonymization via the Cache Side Channel: Attacks and Defenses

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Targeted Deanonymization via the Cache Side Channel: Attacks and Defenses

Authors:

Mojtaba Zaheri, Yossi Oren, and Reza Curtmola, *New Jersey Institute of Technology*

Abstract:

Targeted deanonymization attacks let a malicious website discover whether a website visitor bears a certain public identifier, such as an email address or a Twitter handle. These attacks were previously considered to rely on several assumptions, limiting their practical impact. In this work, we challenge these assumptions and show the attack surface for deanonymization attacks is drastically larger than previously considered. We achieve this by using the cache side channel for our attack, instead of relying on cross-site leaks. This makes our attack oblivious to recently proposed software-based isolation mechanisms, including cross-origin resource policies (CORP), cross-origin opener policies (COOP) and SameSite cookie attribute. We evaluate our attacks on multiple hardware microarchitectures, multiple operating systems and multiple browser versions, including the highly-secure Tor Browser, and demonstrate practical targeted deanonymization attacks on major sites, including Google, Twitter, LinkedIn, TikTok, Facebook, Instagram and Reddit. Our attack runs in less than 3 seconds in most cases, and can be scaled to target an exponentially large amount of users.

To stop these attacks, we present a full-featured defense deployed as a browser extension. To minimize the risk to vulnerable individuals, our defense is already available on the Chrome and Firefox app stores. We have also responsibly disclosed our findings to multiple tech vendors, as well as to the Electronic Frontier Foundation. Finally, we provide guidance to websites and browser vendors, as well as to users who cannot install the extension.

Mojtaba Zaheri, New Jersey Institute of Technology

Yossi Oren, New Jersey Institute of Technology

Reza Curtmola, New Jersey Institute of Technology

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